

Problem Set 9: Solutions

SPEA V706 Math Camp | Module 9 · Thursday, August 13

Part A. Concepts

A1. Covariance from a joint distribution

a. The six probabilities sum to $0.20 + 0.25 + 0.05 + 0.05 + 0.20 + 0.25 = 1.00$, and none is negative. Marginals come from summing across the other variable:

	$Y = 0$	$Y = 1$	$Y = 2$	$\Pr(X = x)$
$X = 0$	0.20	0.25	0.05	0.50
$X = 1$	0.05	0.20	0.25	0.50
$\Pr(Y = y)$	0.25	0.45	0.30	1.00

b. X is a 0/1 variable, so its mean is the proportion of ones: $E[X] = 0.50$. For Y :

$$E[Y] = 0(.25) + 1(.45) + 2(.30) = 0.45 + 0.60 = 1.05$$

c. The product XY is zero whenever $X = 0$ or $Y = 0$, so only the two cells with $X = 1$ and $Y \geq 1$ contribute:

$$E[XY] = (1)(1)(0.20) + (1)(2)(0.25) = 0.20 + 0.50 = 0.70$$

d. By V3:

$$\text{Cov}(X, Y) = 0.70 - (0.50)(1.05) = 0.70 - 0.525 = 0.175$$

Positive, so college graduates tend to get more promotions.

e. X is Bernoulli with $p = 0.5$, so $\text{Var}(X) = (0.5)(0.5) = 0.25$ and $\sigma_X = 0.5$. For Y :

$$E[Y^2] = 0(.25) + 1(.45) + 4(.30) = 1.65, \quad \text{Var}(Y) = 1.65 - 1.05^2 = 0.5475$$

so $\sigma_Y = \sqrt{0.5475} \approx 0.7399$. Then

$$\text{Corr}(X, Y) = \frac{0.175}{(0.5)(0.7399)} \approx 0.473$$

f. Independence would require every cell to equal the product of its marginals. Take the top-left cell: $\Pr(X = 0, Y = 0) = 0.20$, while $\Pr(X = 0)\Pr(Y = 0) = (0.50)(0.25) = 0.125$. One failed cell is enough, so they are **not independent**. The nonzero covariance already implied this, since independence forces the covariance to zero.

A2. The cross term

a. By V5:

$$\begin{aligned}\text{Var}(X + Y) &= 0.25 + 0.5475 + 2(0.175) \\ &= 0.7975 + 0.35 = 1.1475\end{aligned}$$

b. Dropping the cross term leaves $0.25 + 0.5475 = 0.7975$.

c. Ignoring it understates the variance by 0.35, about 30% of the truth. The general rule follows the sign of the covariance: when X and Y move together, the sum swings more widely than the two variances alone suggest, so ignoring a **positive** covariance *understates* the variance of the sum. A negative covariance runs the other way, and the two variables partially cancel, which is the whole idea behind diversifying a portfolio.

A3. Signs, from stories

a. **Positive.** Both rise in hot weather. Temperature drives both, and neither causes the other. Covariance measures co-movement, and it is silent about why.

b. **Negative.** Higher mileage means more wear, so buyers pay less.

c. **Near zero.** There is no plausible mechanism connecting foot size to statistical ability among adults. (In a sample of *children* it would be strongly positive, because age drives both. The population you sample from is part of the claim.)

d. **Near zero, and this is the different one.** Study hours and exam scores are genuinely related, but this sample has been restricted to a narrow band of study hours. With almost no variation in X left, there is almost no covariation to detect, and the correlation collapses toward zero even though the underlying relationship is unchanged. This is **range restriction**, and it is a recurring way to get a misleading zero. A relationship you cannot see is not the same as a relationship that is not there.

A4. Zero covariance without independence

a. By symmetry $E[X] = \frac{1}{5}(-2 - 1 + 0 + 1 + 2) = 0$. And

$$E[Y] = E[X^2] = \frac{1}{5}(4 + 1 + 0 + 1 + 4) = \frac{10}{5} = 2$$

b. $XY = X \cdot X^2 = X^3$, and the cubes are also symmetric about zero:

$$E[XY] = \frac{1}{5}(-8 - 1 + 0 + 1 + 8) = 0$$

c. $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - (0)(2) = 0$.

d. $Y = 4$ exactly when $X = \pm 2$, so $\Pr(Y = 4) = 2/5 = 0.4$. But given $X = 2$, Y must be 4, so $\Pr(Y = 4 \mid X = 2) = 1$. Since $1 \neq 0.4$, conditioning on X changes the distribution of Y , and the two are dependent. As dependent as it gets, in fact: Y is a function of X .

e. Covariance sees only the **linear** part of a relationship, and this relationship is purely quadratic with no linear component. The symmetry of the distribution of X around zero is what does it: the positive and negative halves of X contribute equal and opposite amounts to $E[XY]$, so they cancel exactly. Zero covariance means no linear co-movement. It does not mean no relationship.

A5. Working the rules

a. $\sigma_X = 2$ and $\sigma_Y = 3$, so $\text{Corr}(X, Y) = 3/(2 \cdot 3) = 0.5$.

b. Additive constants drop out of a covariance and multiplicative ones slide out, by V4:

$$\text{Cov}(2X + 3, Y - 1) = 2 \text{Cov}(X, Y) = 2(3) = 6$$

c. Use $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$ with $a = 2$, $b = -1$:

$$\text{Var}(2X - Y) = 4(4) + 1(9) + 2(2)(-1)(3) = 16 + 9 - 12 = 13$$

The cross term enters **negatively** here because of the minus sign on Y . Subtracting a positively correlated variable cancels some of the variation.

d. Projecting Y on X :

$$\frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{3}{4} = 0.75$$

Projecting X on Y :

$$\frac{\text{Cov}(X, Y)}{\text{Var}(Y)} = \frac{3}{9} \approx 0.333$$

They differ because the numerator is symmetric in X and Y but the denominator is not. A slope answers “how much does the outcome move per unit of the regressor,” so swapping which variable is the regressor changes the units of the answer. Correlation, which divides by *both* standard deviations, is symmetric and is the same number either way. Note that the two slopes multiply to $\rho^2 = 0.25$, which is where R^2 comes from in the single-regressor case.

Part B. R

```
pacman::p_load(data.table, ggplot2, fixest, wooldridge)
data(wage1, package = "wooldridge")
dt <- as.data.table(wage1)
```

B1. Covariance and correlation on real data

```
dt[, cov(educ, lwage)]
```

```
[1] 0.6344412
```

```
dt[, cor(educ, lwage)]
```

```
[1] 0.4310528
```

```
dt[, cor(educ, wage)]
```

```
[1] 0.4059033
```

a. $\text{Cov} \approx 0.634$ and $\text{Corr} \approx 0.431$.

b. Neither number's *size* means “weak” or “strong” on its own, and comparing them to each other is meaningless. The covariance is in units of years times log-dollars, so it would change if you measured schooling in months. The correlation is unit-free and bounded in $[-1, 1]$, so 0.43 is interpretable: a moderate positive linear association. Look back at the scatter grid in the M3 deck; 0.43 is a real relationship inside a wide cloud.

c. $\text{Corr}(\text{educ}, \text{wage}) \approx 0.406$, close to but not equal to the 0.431 with `lwage`. Correlation is invariant to *linear* rescaling, and the log is not linear. Taking logs compresses the right tail, which pulls in the high-wage outliers that were dragging the level-scale relationship around. A different transformation is a different variable.

B2. The slope, two ways

```
dt[, cov(educ, lwage) / var(educ)]
```

```
[1] 0.08274437
```

```
feols(lwage ~ educ, data = dt)
```

OLS estimation, Dep. Var.: lwage

Observations: 526

Standard-errors: IID

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.583773	0.097336	5.99751	3.7367e-09 ***
educ	0.082744	0.007567	10.93534	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.479165 Adj. R2: 0.184253

a, b, c. Both give 0.082744. They must agree, because the OLS slope *is* the sample analog of $\text{Cov}(X, Y) / \text{Var}(X)$. `feols` is doing the same arithmetic with more machinery around it. Note that `cov()` and `var()` both use the $n - 1$ divisor, so it cancels in the ratio.

d. One more year of schooling is associated with about 0.083 higher log wages, or roughly an 8.3% higher wage. (The exact figure is $e^{0.083} - 1 \approx 8.6\%$; the log-point shortcut is close for small values and drifts as the coefficient grows.) “Associated with,” not “causes.” People who get more schooling differ in other ways, and untangling that is what the rest of V706 is about.

B3. Recover a correlation you built

```
sim_cor <- function(n, rho, seed = 20260813) {
  set.seed(seed)
  x <- rnorm(n)
  e <- rnorm(n)
  y <- rho * x + sqrt(1 - rho^2) * e
  cor(x, y)
}
```

```
c(n50 = sim_cor(50, 0.6),
  n500 = sim_cor(500, 0.6),
  n50000 = sim_cor(50000, 0.6))
```

n50	n500	n50000
0.5492573	0.6087477	0.6015364

a. At $n = 500$ the estimate is about 0.609, close to the true 0.6.

b. At $n = 50$ it lands near 0.549, off by about five hundredths. At $n = 50,000$ it is about 0.6015, off by about one thousandth. More data, tighter estimate. Run it with different seeds and you will see the small- n estimate bounce around a lot while the large- n one barely moves.

c. Estimates computed from samples are themselves random, and their spread shrinks as the sample grows. That is exactly the law of large numbers and the σ^2/n sampling variance you meet tomorrow morning.

B4. Correlation is blind to curvature

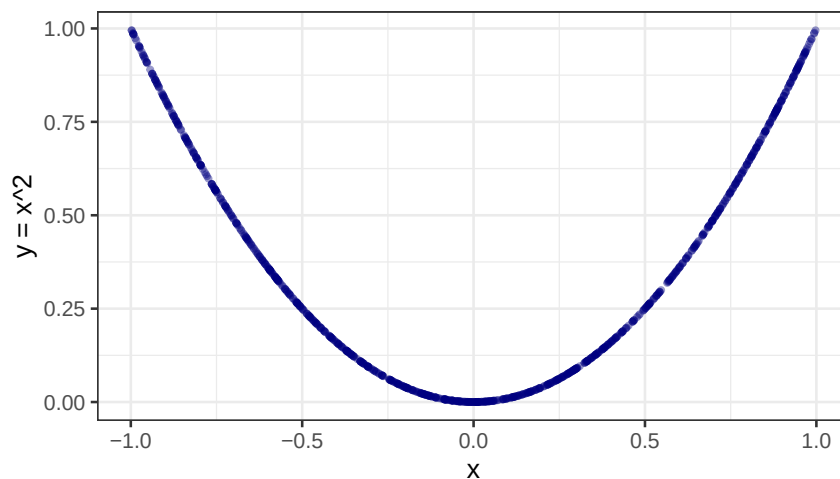
```
set.seed(20260813)
x2 <- runif(1000, min = -1, max = 1)
y2 <- x2^2
cor(x2, y2)
```

```
[1] 0.05127389
```

a. About 0.051, which is zero for practical purposes. The true population correlation here is exactly zero; the small residue is sampling noise, the same phenomenon as B3.

b, c. The plot is a clean parabola. Every point lies exactly on a curve, so knowing x_2 tells you y_2 with no error at all:

```
ggplot(data.table(x2, y2), aes(x2, y2)) +
  geom_point(colour = "#000080", alpha = 0.4, size = 0.8) +
  labs(x = "x", y = "y = x^2") +
  theme_bw(base_size = 10)
```



The correlation is near zero and the dependence is perfect. Correlation summarizes how well a **straight line** describes the cloud, and no straight line describes a symmetric parabola. This is A4 with 1,000 simulated points instead of five hand-computed ones. Always look at the scatter plot.

B5. The same analysis, three ways

a. The console. Nothing survives. Reproducing the slope means retyping every command from memory, and after a restart you cannot even scroll back to find them reliably. The console is for looking around, never for producing a number you intend to report.

b. One script. The script runs from a clean session as long as everything it needs is written in it. The classic failure is having created an object in the console (a recoded variable, a filtered table) and never adding that line to the script. It works all afternoon, and it breaks the moment you restart. Restarting R and re-running is the only honest test, and it is worth doing before you send anyone results.

c. A pipeline. Something like:

```
# main.r
library(here)
source(here("code", "01_prepare.R"))
source(here("code", "02_analyze.R"))
```

`here()` resolves paths from the project root, so the code runs the same on your laptop, on a coauthor's machine, and on a cluster, with no `setwd("/Users/me/Desktop/...")` line to break.

Which one to use. The pipeline, for anything you will revise. Two years from now you will not remember which script produced which table, and a referee will ask you to change one sample restriction and rebuild everything downstream. A pipeline makes that one edit in `01_prepare.R` followed by one run of `main.r`, and it makes the dependency between steps explicit rather than remembered. The cost is about ten minutes of setup, paid once.

Check yourself.

1. Scale. Covariance is in the product of the two variables' units, so its magnitude depends on how you happened to measure things. Correlation divides that out and lands in $[-1, 1]$, so the number is comparable across variables and across studies.
2. X uniform on $\{-2, -1, 0, 1, 2\}$ with $Y = X^2$, from A4. Any symmetric distribution paired with an even function of it will do.
3. $\text{Cov}(2X, Y) = 2 \text{Cov}(X, Y)$, so the covariance doubles. The correlation is **unchanged**, because $\sigma_{2X} = 2\sigma_X$ doubles the denominator too. Correlation is invariant to linear rescaling, which is exactly why we report it.